At the beginning of this century, most scientists unanimously agreed that an airplane was probably impossible, and, even if it worked, it was impractical. The eminent American astronomer Simon Newcomb declared with finality:

> *The demonstration that no possible combination of known substances, known forms of machinery and known forms of force, can be united in a practical machine by which man shall fly long distances through the air, seems to the writer as complete as it is possible for the demonstration of any physical fact to be.*

Fortunately, the Wright brothers did not have time to worry about Newcomb's conclusions. They were too busy bolting a gasoline engine onto some wings in their bicycle shop in Dayton. Here's what William H. Pickering, a well-known scientist, had to say after the Wright brothers had flown their airplane at Kitty Hawk:

> *The popular mind often pictures gigantic flying machines speeding across the Atlantic and carrying innumerable passengers in a way analogous to our modern steamships. ... It seems safe to say that such ideas must be wholly visionary, and even if a machine could get across with one or two passengers the expense would be prohibitive to any but the capitalist who could own his own yacht.*
>
> *Another popular fallacy is to expect enormous speed to be obtained. It must be remembered that the resistance of the air increases as the square of the speed and the work as the cube.... If with 30 h.p. we can now attain a speed of 40 m.p.h., then in order to reach a speed of 100 m.p.h. we must use a motor capable of 470 h.p. ... it is clear that with our present devices there is no hope of competing for racing speed with either our locomotives or our automobiles.*

Scientists may have struck out when it came to airplanes, but they had another chance to improve their batting average on predictions as the rocket age approached. In spite of the pioneering researches of American Robert Goddard and the Romanian Hermann Oberth, who outlined in detail the basic technology of rockets and spaceships, Professor A. W. Bickerton in 1926 wrote:

> *This foolish idea of shooting at the moon is an example of the absurd length to which vicious specialization will carry scientists working in thought-tight compartments. Let us critically examine the proposal. For a projectile entirely to escape the gravitation of the earth, it needs a velocity of 7 miles a second. The thermal energy of a gramme at this speed is 15,180 calories. . . . The energy of our most violent explosive—nitroglycerine—is less than 1,500 calories per gramme. Consequently, even had the explosive nothing to carry, it has only one-tenth of the energy necessary to escape the earth. . . . Hence the proposition appears to be basically impossible. . . .*

During the last few months of World War II, the Germans surprised the world with a V-2 rocket, which they fired from the Continent to England. This naturally raised the possibility that an intercontinental missile might be built which could he fired in Europe to destroy American cities. Dr. Vannevar Bush, who was head of the United States scientific war effort, testified before a Senate Committee on December 3, 1945:

> *There has been a great deal said about a 3,000 miles high-angle rocket. In my opinion such a thing is impossible for many years. The people who have been writing these things that annoy me, have been talking about a 3,00U mile high-angle rocket shot from one continent to another, carrying an atomic bomb and so directed as to be a precise weapon which would land exactly on a certain target, such as a city. I say, technically, I don't think anyone in the world knows how to do such a thing, and I feel confident that it will not be done for a very long period of time to come. ... I think we can leave that out of our thinking. I wish the American public would leave that out of their thinking.*

Slightly over a decade after this expert delivered his words of wisdom, there were intercontinental missiles in actual production, and the Russians had Sputnik I orbiting the earth ! Arthur C. Clarke in his excellent book Profiles of the Future (and to whom we are indebted for the examples of predictions given in this chapter) wrote:

> *Too great a burden of knowledge can clog the wheels of imagination; I have tried to embody this fact of observation in Clarke's Law, which may be formulated as follows:*
>
> *When a distinguished but elderly scientist states that something is possible, he is almost certainly right. When he states that something is impossible, he is very probably wrong.*
>
> *Perhaps the adjective "elderly" requires definition. In physics, mathematics, and astronautics it means over thirty; in the other disciplines, senile decay is sometimes postponed to the forties. There are, of course, glorious exceptions; but as every researcher just out of college knows, scientists of over fifty are good for nothing but board meetings, and should at all costs be kept out of the laboratory!**
>
> ** Arthur C. Clarke, Profiles of the Future (New York: Harper & Row 1964), p. 14.*

Men Who Scored

There have been many men who have been successful in making remarkably accurate predictions of the future. Leonardo da Vinci possessed the necessary combination of imagination and nerve. Jules Verne, in the last century, gave us fantastic predictions, almost all of which have become present-day realities. Thorsten Veblen was able to foresee economic and social trends far in advance of their occurrence. H. G. Wells encompassed, at least in outline, the inevitability of a world society essentially based upon

scientific, rather than political, orientation.

Anyone who doubts the possibility of long-range prediction might well consider the statements of Friar Roger Bacon, who lived between 1214 and 1294. These words were written at a time when science and technology as we know them were non-existent :

> *Instruments may be made by which the largest ships, with only one man guiding them, will be carried with greater velocity than if they were full of sailors. Chariots may be constructed that will move with incredible rapidity without the help of animals. Instruments of flying may be formed in which a man,sitting at his ease and meditating in any subject, may beat the air with his artificial wings after the manner of birds ... as also machines which will enable men to walk at the bottom of the seas. . . .*

Perhaps the only thing we can be sure about when predicting the future is that it will sound utterly fantastic. It will be enormously different from anything that we regard as "natural" or "right." If our predictions in this book seem plausible to you, we have probably failed to see far enough ahead. If our projection of the future seems completely impossible and utterly fantastic, there is a possibility that we may be on the right track.

How We Predict the Future

As we pointed out in the first chapter, we have not been able to find a crystal ball we consider reliable for predicting the future. We have, instead, developed a method of analysis which we wish to set forth clearly. If you agree with the method by which we will attempt to predict some of the features of our twenty-first century civilization, perhaps you will then find some of the revolutionary conclusions more acceptable.

It is our hypothesis that there are three important factors that will greatly influence the evolution of our civilization. They are:

1. The values, purposes, and ideals toward which man is striving.
2. The method of thinking that we use to select our courses of action.
3. The state of technology, or, what sort of tools are available to help us do what we want to do.

In the next three chapters we are going to discuss each of these three factors that we believe will generate the future shape of our civilization. We believe that if we are able to determine what people really want here on earth, we will be able to anticipate the general directions in which men will go in building the civilization of the future. We will inquire into the "mind" and "heart" of man to see what he really seems to want most. We will see if there have been values that have been on a steady upswing for centuries. If we are successful in correctly pinpointing the values and ideals that human beings will most

desire in the future, we will be on our way toward accurately predicting the type of civilization that lies ahead of us.

Values are only a first step. To know what man really wants is most helpful, but it is equally important that we accurately pick the method of thinking that he will use to try to get what he wants. For example, man has always placed a high value on good health. But simply having this value is not enough. The methods of thinking that accompany a value have a lot to do with whether it will be achieved. Thinking methodology that was in vogue during past centuries resulted in seriously ill people being bled by barbers. Blood-sucking leeches were considered a necessary item in the doctor's bag. In contrast, the methods of thinking in use today in the field of medicine often call for a blood transfusion.

In the past if an idea sounded plausible, people believed it. If an authority said something was true, it was generally accepted. It was very rare for a person to say, "I don't care how reasonable it sounds or who says it's so, I want to make some careful tests to see for myself."

Thus, the methods of thinking that people employ play a big part in determining what kind of civilization they have.

A third factor that interacts with the value structure and the methods of thinking is the technology of a given age. For example, as a consequence of our value structure, we want to travel to the moon. We use scientific methods of thinking to arrive at the basic theories that show us how to get there. But if the state of technology does not yield a metal strong enough and light enough to build spacecraft, we're not likely to set foot on the moon.

Our technology has been slowly evolving for the last 600,000 years. It was a great day when prehistoric man first picked up a stick and used it as an instrument to achieve greater control over his environment. The invention of the wheel was a big step forward. But there has been more technological development in the last half-century than in the preceding 600,000 years. Computers, automation, and the development of atomic power enormously enhance man's potential to achieve whatever values and ideals he chooses.

Now you know how we are going about this. If anyone wishes to spell out in useful detail some of the forms of our future, we believe he must pick the right horse in three different races:

1. He must correctly assess what man will want to do—what he really values most.
2. He must accurately find out how he's going to try to do it— what methods of thinking he will rely upon most.
3. He must analyze the tools that man will have for accomplishing what he sets out to do—he must pinpoint the significant technological developments that will play major roles in the future.

All three factors interact with each other. The value structure not only influences the method of thinking and the technology, but it is, in turn, influenced by them. The method of thinking that man employs is affected by his value structure and the technology of the age, but it also plays a part in modifying both of these. Similarly, the technology of any given civilization interacts in a mutual way with the value structure and the method of thinking. These pregnant factors might be viewed as three gears that mesh with each other.



Now you have a way to judge how well we accomplish what we set out to do. If you don't feel that we have picked the trends that will play the largest part in shaping the future, then you probably won't accept the type of civilization that we describe in Part II of this book. If, however, you feel that we have correctly assessed the dominant trends in mankind's value system, if you feel that the method of thinking has been accurately pinpointed, and if you view technology in the same vein, then you may be intrigued by our projection of the experiences of a man and woman in the twenty-first century.

4.Our Values Chart Our Course

Many people have pessimistic feelings about the future. Perhaps the enormous growth of technology that enables man to build bombs that can destroy a city, automobiles that kill more people than wars, and planes that exceed the speed of sound give them a "What's next?" feeling. Perhaps some of the pessimism is due to the failure of economic, social, and political inventions to keep pace with the developments of physical science.

Many people identify the future with the value system projected by Aldous Huxley in *Brave New World* and George Orwell in *1984.* These authors had an important message for us. They pointed out that if we choose values that ignore the inner needs of man, we can create a horror here on earth. Orwell wrote of a society in which technology was used to the ultimate to curtail freedom of thought and action. A television camera in every apartment enabled secret police to see what was going on. The only way for an individual to have any privacy was to turn his back to the camera and speak softly. This was modern technology implementing a value system of the past. Most science fiction uses an inadequate system of crude values that conflict with man's need to be a free spirit enjoying the potentialities of life.

We are rapidly approaching a stage in human development that will permit man to do almost anything he wants. We feel confident man will choose a set of values that will yield happiness. The warnings of Orwell and Huxley have not been in vain. We'd be glad to trade two hours in the present world of conflict for one hour in the advanced civilization of one hundred years from now. We feel that the next century will open the door to new horizons of human experience and happiness.

How We Got Our Value System

Perhaps the most significant thing a person can know about himself is to understand his own system of values. Almost every thing we do is a reflection of our own personal value system. What do we mean by values? Our values are what we want out of life. No one is born with a set of values. Except for our basic physiological needs, such as air, water, and food, most of our values are acquired after birth.

As an example of how values are picked up, individuals who grow up in the twentieth-century America are conditioned to the acquisition of money as a part of their value structure. It isn't "natural" for a person to Want money. It's a value most people in our culture acquire. When we were perhaps two years old, we learned that a penny would buy a piece of candy. We heard adults say with obvious approval, "He has a lot of money." As we grew up, we observed the correlation between money and larger cars, more beautiful homes, finer clothes, etc. Feelings of self-worth became associated with money. Over the years we gradually acquired a value structure that led us to place such a high value on money that in our present economy of scarcity we spend a large part of our lives scratching for dollars.

If we had grown up in a society where people do not stress material wealth, money would not have played an important part in our value structure. In Samoa food is there for the taking. Coconuts and fruit grow without cultivation, fish are available in the nearby sea, little or no clothing is needed, and it doesn't take long to put up a thatched-roof hut from materials that grow on all sides. In such a society people may be less likely to work hard for money. People conditioned in this manner do not act in ways men consider acquisitive, competitive, and thrifty.

When a culture is in a rapid state of transition, we have conflicting values which often yield hostility and unhappiness. Most people in twentieth-century America have a value system that includes both sincerity and the desire for money. Suppose a merchant advertising a sale were to include a "loss-leader" item such as a dacron shirt for $1.49. But if people flocked to purchase this advertised shirt, they would find that he had only six for sale at this price. The conflict in values might be felt by both the merchant and his customers. Our transitional culture of today keeps us in perpetual conflict.

Values are not immutable and eternal. They are created largely by the feelings of human beings. "The values both of individuals and of groups both large and small have been changed," Ralph Borsodi advises us:

> *They are being changed today more rapidly than ever before. The acquisition and the inculcation of values begins at birth in the acceptance or rejection of the child by its parents;the values inculcated and acquired during the first few years of life affect the unconscious mind so deeply that their alteration is not easy, but they can be changed and it is easier to change them in the modern . . . world than ever before.**
>
> ** Ralph Borsodi, "Eight Propositions About Values," The Humanist, (1964, Number Five), p. 152.*
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We can test a value by asking, "Does it yield happiness for us?" There is nothing absolute about any value—it's not a question of "right" or "wrong," "true" or "false," "good" or "bad." We should ask ourselves, "How do our values affect our happiness, both present and future?"

The happenstance of when and where we are born determines what set of values we acquire. A few years or a few miles can make a whale of a difference. Although we know that no one is born with a set of values, the particular set of values we acquire seems both "inborn" and "natural" to us. We always feel strange when we meet people with a different set of values. They don't act "right."

Jacques Fresco, one of the authors, visited Bora-Bora in the Society Islands thirty years ago. He carried with him fifty pocket mirrors and hundreds of beads and other novelties that he hoped to use in place of money. Fresco showed these gifts to several natives and then put them into a suitcase. When he returned to his living quarters several hours later, he was surprised to see a number of men and women freely handing out the beads, mirrors, and other trinkets to their Polynesian friends. He interrupted them and asked what was going on. One of the older men replied that Fresco had more than he needed and that it was a shame to see so many wonderful things not being put to use.

After establishing a favorable relationship with the Polynesians, he made a casual comment that he would appreciate their help in building an outrigger canoe. They gathered in a huddle and left without saying anything. Several days later, they appeared carrying an outrigger canoe and presented him with this extravagant gift. During the next two weeks he was unable to find time to enjoy their gift. They appeared disturbed and picked up the canoe and carried it away.

"Why are you taking my canoe away?" he asked.
An older Polynesian stepped forth and commented resentfully, *"We give you boat. You no use. We take back."*

It took Fresco a while to grasp fully the significance of such a system of values based on need and use, instead of ownership.
Many of the values that you and I have today were devised thousands of years ago in a world of great scarcity. For example, during past centuries it took about twenty farmers to produce enough surplus food to maintain one townsman. To have a deep feeling of worth in such a world, it was usual to have a value structure that generated hardworking and thrifty behavior.

The scarcity conditions of the past have led men to place a high value on owning as many things as possible. The ownership of a set of tools enabled a man to make his living. If someone had stolen these from him, his ability to earn a living would have been threatened. The ownership of material goods became associated with a feeling of self-esteem. In some of our past cultures, the concept of private ownership even extended to include women. Women could be bought and sold in the marketplace. Even in the twentieth century we have conflicts in our value system as the status of women is evolving from being the property of men to the role of free human beings.

"Life, Liberty, and the Pursuit of Happiness"

If we are to make a reasonable anticipation of the type of world our descendants will have, we must adequately choose the dominant trends in values. So here we go! Although there are many ways to phrase it, we feel that the value wave of the future was expressed well by Thomas Jefferson in The Declaration of Independence in that historic phrase, "Life, Liberty, and the Pursuit of Happiness."

Although we in mid-twentieth-century America have a degree of liberty and freedom, we have only reached the first rung in climbing the ladder toward achieving the highest degree of life and liberty. When men of the future look back and try to understand the relatively primitive conditions of mid-twentieth-century America, they will be most perplexed. They may comment that while Fourth of July speeches in which the blessings of liberty, freedom, and the right of everyone to be an individual were spoken of quite highly, yet every year more and more laws were passed telling people that they couldn't do this, and they would pay a fine if they did that, and they would go to jail for doing something else. Perhaps further research by future historians might reveal that many of our laws were passed because we felt they were needed to keep men from hurting other men. In other words, the liberty of one person had to be limited so that he could not destroy the freedom and happiness of others. And this will be a curious thing for people in the future to understand about our present civilization. For they will find it unthinkable that conditions existed that permitted conflict between human beings.

The civilization of the future will outgrow the need for laws as we know them. For example, we have a law against murder.

In the future there will be no laws dealing with murder. No baby is born a murderer. Immersed in a conflict-culture such as ours, he interacts with the conditions of his environment to develop a pattern of reaction that can, under certain circumstances, lead him to kill another human being. As he grows up, he often sees headlines and pictures in newspapers telling of murder. In literature, movies, and television he witnesses thousands of murders.

Our sick society dotes over such legendary murderers as Jesse James, Al Capone and Bonnie and Clyde. A young child lives under conditions where he is trained to be jealous and acquisitive. His king-sized ego learns to respond with feelings of deep hurt and rejection. If one day the grown man were to find his wife in bed with another man, he would respond in a way that is dictated by the years of conditioning. It would seem natural for him to seize a gun and murder the man.

Today we are beginning to identify various things which condition us to act as we do. In the future the factors that condition human beings to kill or do other things that harm fellow human beings will be understood and eliminated. The value structure will not permit children to be conditioned in sick, twisted, and insane ways. If it is found that someone might do something that could hurt another human being, the reaction will not be to pass a law against it. People will not use an archaic structure of courts, judges, and laws. They will simply ask themselves, "What is it that permits a person to act in a way that could hurt someone else?" When they find the root of the difficulty, they will modify conditions so that people will not

—or can not—act in this way.

What About Human Nature?

When little was known about cultural anthropology, sociology, and psychology, it seemed quite valid to resist proposed reforms by saying, "It won't work. It's against human nature." It is difficult for many people to appreciate the fact that what they call "human nature" just doesn't exist. Scientific research has discovered that it is probably not even "human nature" for a man to be attracted sexually to a woman! The particular object to which one is attracted sexually seems to be determined by experiences that happen early in life. This even applies to some animals. Eckhard Hess kept a young male jungle fowl with him for the first month of its life. It was not permitted to be with any of its own species during this time. Hess noticed :

> *This animal, even after five years—much of that time in association with his own species—courts human beings with typical behavior, but not females of his own species. This certainly is a far-reaching effect and is similar to the finding of Räber (1948), who reported on a male turkey whose behavior toward human beings was similar.**
>
> ** Eckhard H. Hess, "Imprinting," Science, (1959), p. 140.*

Man is like a mirror—he largely reflects his surroundings. If man were to come into the world with a fixed "nature" consisting of automatic responses, civilization would be impossible. Like the ants, we would live out our lives in patterns that are modified but little with the passing of time. The wonderful thing about us is that we come into this world with maximum flexibility.

Values of the Future

We believe that man's pursuit of happiness under the conditions of the twenty-first century will permit him to achieve the fullest expression of the following values:

Life and Liberty
A matured society will permit man a maximum degree of life, liberty, and freedom. Individuals will understand themselves. They will choose patterns of living that will deeply express their own inner selves. Never before has society been able to permit all individuals to express their intellectual, emotional, and physical needs. At long last, individuals will no longer be subordinated and pressured to conform to a set pattern.

Economic Abundance.
There will be economic abundance so that the material needs of men are amply met. Competitiveness, acquisitiveness, thriftiness, and the hardworking syndrome will be as extinct as the dinosaur.

Health and Longevity.
The ideal of radiant health will continue to be valued as it has in the past. But for the first time it can be realized. Scientific nutrition along the lines suggested by one of the authors in How to Live Longer—Stronger—Slimmer will be built into everyday life-patterns.* Improved genetic designs and living conditions which promote maximal health will give a freedom from disease and a maximum of energy that was seldom achieved in older civilizations. The average life-span may well exceed 150 years in the twenty-first century. Eventually, it will stretch toward immortality.

* Kenneth S. Keyes, Jr., How to Live Longer—Stronger—Slimmer (New York: Frederick Fell, 1966).

Love and Friendship.
Man's feelings of friendship, warmth, and love for all other men will deepen to an extent that can not be understood by those who live in the twentieth-century world of scarcity. Friendship and love in the twentieth century are so clouded with conscious and unconscious hostilities, competitiveness, envy, greed, and insecurity that the deepest levels of human warmth can not be approached. Only in a mature society is it possible for man to savor fully his relationships with fellow human beings.

Physical Pleasures.
The potential that men and women have for enjoying the pleasures of sex will reach their greatest fruition in a mature society. Sexual behavior in mid-twentieth-century America will be regarded as incredibly primitive, for it is overladen with guilt feelings inculcated by early raining. In the twenty-first century sexual emotions will be treasured because of their historical association with the creation of life. Deep feelings of pleasure, oneness, and relaxation will flow from the mature expressions of human sexual feelings.

Appreciation of Beauty.
Man's appreciation of beauty will expand from the narrow ranges of the present into the greatly enlarged horizons he will achieve in the future. The physical beauty of human beings will not be confined to the narrow standards of the "beauty queen" mentality of today. The beauty of human beings of all ages from birth to the lovely mellowness of old age will be appreciated. Esthetic experience will become a pervasive reality in the lives of all men. Almost everything in the twenty-first-century world will be beautiful. Man's esthetic sense will not be dulled by exposure to sham and artificiality, slums, jukeboxes, and advertising art. As we will discuss in a later chapter, music will acquire new dimensions that completely transcend the limited orchestral ranges of today. Beauty will become an integral part of life, not just something we appreciate at detached moments. People will be more interested in producing art than in acquiring and displaying it.

Deep Levels of Self-Knowledge and Communication of Feelings.
People in the twenty-first-century world will achieve penetrating levels of rapport, both with themselves and with the feelings of others. Many of the inner feelings of people in the twentieth century are repressed and do not come fully into awareness. It is extremely rare that one's innermost feelings can be continually, fully, and freely expressed, even between friends or lovers. In the future all feelings will be eagerly sought, verbalized, and thoroughly accepted by others. This will produce a new dimension of relaxed living that is

almost unknown today.

Vicarious Sharing of the Delights of Others
The relaxed egos of individuals in the twenty-first century will permit them to achieve a deep pleasure in sharing the happiness and the experiences of others. The achievement of happiness will not be on a narrow self-centered basis in which one ego fights another for a feeling of worth. People in the future will feel that the happiness levels of all individuals are to a large degree interlocked and rise and fall together. For example, if one person is sick, the disease might spread to others. If he is unhappy, the interaction with others might make them unhappy. If an individual is given an inferior status, his resentment might lead to hostile acts that hurt others. Hence, individuals in the twenty-first century will value the feelings of other people as their own and derive a deep satisfaction from knowing that all men in their society live relaxed, deeply-fulfilled and fascinating lives. No one will stand alone.

The Challenge of Life.
The challenge of life that men and women will experience in the future will, perhaps, be a supreme value. For the first time all men and women will live a multidimensional life, limited only by their imagination. In the twentieth century we could classify people by saying, "He is good in sports. She is an intellectual. He is an artist." In the future all people will have the time and the facilities to accept the fantastic variety of challenges that life offers them. Men and women will feel perfectly at home in all parts of their world. The satisfaction of a continuing self-development will be a normal part of life, not a rare thing as it is in the mid-twentieth century.

Open Eyes and Open Minds

As we sketch a picture of the twenty-first-century world, we will seek values that will help individual men, women, and children to achieve feelings of fulfillment. Most of the functional value systems of the past will seem inappropriate in the world of the future that we project in Part II.

We must be prepared to see the dissolution of human institutions that have been with us for thousands of years if they no longer contribute maximally to human happiness in the changed world of the twenty-first century. As we study the future value structures of mankind, we must not be like travelers who go to a foreign land and immediately compare everything with their own home town. To understand another place, we must lay aside the value patterns that we are used to. We must relax our mental sets so that we may feel a new pattern of human experience. The biggest problem that we face is to get the twentieth-century dust out of our eyes so that we may feel and think as freshly as possible about the almost limitless permutations and combinations of life patterns that mankind may explore for ever higher levels of fulfillment in the future.

If you think that today's vices and virtues are absolute and ultimate and reflect the final value system for all times and all civilizations, you will find our projection of the future to be shocking and incredible. If you have an absolute attitude toward values, all you can do is to project your particular conditioning onto mankind's dynamically evolving future. You will tend to see the future in terms of the present with, of

course, some of the burrs removed. If you want even the slightest chance of understanding where we are going—and possibly helping us get there—shake out the absolutes and put in the relatives. A culture must be seen relative to time, relative to place, and relative to a particular framework of values, thinking methodology, and technology.

Civilization has just recently given up crawling and has begun to toddle. With the development of scientific methods of thinking several centuries ago, mankind began to blossom into what might be called the childhood of civilization. We are today fast-growing adolescents. We have clashing values. We are torn between our inside feelings and needs and outside structures and pressures. The adulthood of human civilization lies before us. There is only one thing we can know for certain—the world of the future will be enormously different from anything in the past or present.

5.The Scientific Method

Our method of thinking helps us choose between formulations, ideas, thoughts, notions, hypotheses, theories, and other cerebral itches. It enables us to decide what is "true" and what is "false." Our method of thinking should help us pick the most reliable thing to do that offers maximum predictability. It should enable us to reject ideas that do not correspond with observable facts.

What are the various methods of thinking that man has used? There is the method of appealing to authority, or, asking what the wise men, present or past, have to say about the problem. There is the method of intuition, which means pumping your feelings for something that may bear on the problem. There is the method of rational, philosophical, logical thinking, which means using your brain to test out various verbal structures.

We're in favor of using all the above methods of thinking, and any others you can find, for the purpose of coming up with creative ideas that may be useful. It is vital that we should not misuse these methods of thinking by relying on them to make a final selection. To choose the most useful ideas we must finally quit talking and check whether a verbal formulation corresponds with observable facts.

The history of human thought shows that we don't get very far as long as we spin words around in our heads and fail to take the scientific step of checking them against observable facts. Non-scientific methods of thinking do not produce agreement between individuals of different backgrounds. They can argue "until the cows come home," and problems still don't get settled. Even worse, non-scientific methods have never been successful in building an effective structure of knowledge on which men of all nations can rely. Reliable knowledge accumulates only when men slow the flow of words and start scientifically checking their ideas against something outside their skulls. The enormous progress of science and the technological marvels of our age were possible because men tested their ideas against observable facts.

The Beginning of the Scientific Method

Although the Greeks dabbled in and seemed to have anticipated almost everything, it was not until Francis Bacon (1561-1626) appeared on the scene that men began to use systemically the method of science. As Dr. George Gallup pointed out :

> *Bacon argued strenuously for an entirely new approach to the physical world. ... He advocated the experimental approach, its virtue being that every finding and conclusion could be tested. Since every new bit of knowledge was demonstrably true, there would be no room for quarreling schools of thought, and knowledge about the physical world could be rapidly expanded. Bacon's foresight has proved correct.*
> *The world of knowledge, as a result, has been literally transformed by this special way of channeling man's intelligence.*
> *Almost all of the physical and material benefits mankind enjoys today are a product of this method of science.**
>
> * George Gallup, The Miracle Ahead (New York, Evanston, and London: Harper & Row, Publishers, 1964), p. 153.

In a way this scientific method of thinking is really nothing new and not so unusual. We use it often in our everyday personal, business, and social problems. The thing that makes the big difference is the thorough insistence that all knowledge pass the test of being checked by observation. Most of our everyday thinking is a mixture of all methods of thinking.

The scientific method is almost as "old as the hills." Imagine a group of cave men perched on a river bank, arguing whether Onk or Donk can run faster. They'll probably use all of the methods of thinking that we have described above. They may get the opinion of the wise men of the tribe (reference to authority). They'll probably argue and argue (the use of logic). Onk has bigger muscles, but Donk has longer legs, etc. The words will whirl around endlessly. Perhaps one of the women will use her intuition and try to pick the man that can run faster. Or they may use a method to settle the argument that, unfortunately, is still with us—they may fight about it. The one who beats up the other one is assumed to be right. This is the way of the jungle, and it is universally relied upon by lions, tigers, wolves, etc. Unfortunately, in our mid twentieth-century world almost all really big disputes between nations are settled in this age-old fashion.

None of the above methods of settling the dispute may be called scientific. The words go 'round and 'round, the tempers go up and up, but all this has very little to do with who can run faster. Finally, some little genius perched on a branch interrupts the sweating people below, "Why don't we let them run to the big oak tree over there and see who is faster?" This genius has proposed a method of settling the dispute that would probably enable most reasonable men to reach agreement. It is the method of science. It is a method that does not involve the use of words to decide the problem.

The two men get ready to run. The signal is given. They start running. Everybody stops arguing and starts observing. They are anxious to know the facts. They use their senses to report information that will help them decide. They will carefully note whether Onk arrives first, Donk arrives first, or whether they both get there at the same time. The argument as to which could run faster is settled at that time by using the scientific method.

If these cave men had generalized the lesson and had decided to use observation to test all their arguments, problems, ideas, and theories, the human race could have developed today's civilization about fifty thousand years ago! They could have fought out all of the atomic wars and gone through the uncomfortable transition periods a long time before we came on the scene to sweat it out. But instead we find that man became too intrigued with verbal intricacies and developed the habit of using verbal means to decide between "true" and "false."

The Greek philosopher Aristotle had one of the most brilliant minds the world has ever produced. According to some ancient authors, he wrote 1,000 volumes that covered practically every field of learning. When Aristotle was writing on physics, he explained that a heavy object would fall faster than a light object of the same shape and material. This verbal conclusion seemed so natural and obvious to him! How silly to bother with testing. He just reasoned it out in a way that seemed logical. Weight makes things fall. Therefore, the more weight, the faster the fall. Seems reasonable, doesn't it? Aristotle had many assistants, and it would have been very easy for him to check this by actually dropping a light rock and a heavy rock from the top of the Parthenon. But he did not depend on the scientific method for checking his thinking. He liked to prove things rationally, logically, intellectually. He didn't know it was necessary to test the results of his brilliant mind by observation.

You and I lost two thousand years of progress because of this non-scientific attitude of Aristotle's age. In the sixteenth century Galileo began to wonder about the predictability of Aristotle's theory of falling objects. But he didn't refute Aristotle with intellectual argument. He used the scientific method for checking it out. He let the facts speak for themselves. Instead of "settling" the problem by words, he dropped a heavy weight and a light weight at the same time. They both hit the ground simultaneously. "Nature" had spoken. Sane people stopped arguing about the problem. Regardless of how illogical it might seem to us, the facts are that the weight of an object, under standard conditions, does not determine the rate at which it falls.

When we test our ideas by our senses, we can go forward. We can build useful structures of reliable knowledge. We can predict. The unscientific tendency to believe without testing is not, alas, confined to the Ancient Greeks. "Men are apt to be much more influenced by words," said the famous scientist Pavlov, "then by the actual facts of the surrounding reality."

Scientific Methodology

Scientific thinking may be boiled down to three steps:

Coming Up with New Ideas.
We use our imagination, intuition, memory, etc. to suggest ways of explaining or solving a problem.

Processing Ideas Mentally.
We analyze our thoughts to determine a way to test them by observation. By logic we can determine how well our brain child fits in with known facts. We can try to figure out intellectually how well it will work. What will be its consequences?

Testing by Observation.
After we have found a possible new solution to our problem, explored it, and figured out a way to check it, we are then ready to take the third step which will make our thinking scientific. This is testing by observation. This is where we shut up and let observable facts do the talking.

The Scientific Method in Action

Suppose you were to get into an argument with someone about whether a chameleon will change its color to match its surroundings. Perhaps you have heard the tale of a chameleon that went crazy trying to match a Scotch plaid. Now, if you would want to pursue this matter in an argumentative or philosophical fashion, you could just stand around whirling the words all day and night. But if you felt like getting useful information on the matter, then it would be time to stop talking and start doing. You would bide your time until you could round up one or more chameleons. Then you would let them demonstrate their alleged powers of matching their background. People who have the scientific habit of letting the facts speak for themselves have observed that a chameleon can no more pick its color than a zebra can pick its stripes. When these people check things out, they find that a chameleon turns green when it is excited, frightened, angered, asleep, or dead. It turns brown in response to low temperatures, hunger, and strong sunlight. If you take a green chameleon and throw it into the icebox, it will take about three minutes for it to change from green to brown.

If an argumentative person were to use his logical ability and debate the subject of how chameleons should act, it would be hard to convince him that chameleons don't give a hoot about matching their backgrounds. If you could persuade him to check his thoughts by observation, he would find that a green lizard is perfectly comfortable on a brown background and a brown lizard is perfectly at home on a green background. He'll even let you photograph him that way.

Anyone who tests his thoughts by using his eyes, ears, touch, smell, or taste is using the method of science. "The scientific method," says Stuart Chase:

> *is concerned with how things do happen, not how they ought to happen. Knowledge of the way things do happen, with no ifs, ands, or buts, allows us to deal more effectively with our environment. The method is no more an exclusive matter for professionals than it is a matter of white coats and goggles. Most of us are amateur scientists today, though we are seldom aware of it. . . . The scientific method is not primarily a matter of laboratories and atom-smashers or even*

*meter sticks; it is a way of looking at things, a way of gathering from the world outside knowledge which will stay put, and not go wandering off like the wickets in Alice's croquet game.**

** Stuart Chase, Tyranny of Words (New York: Harcourt Brace and Company, 1938), pp. 123-24. Reprinted by permission of Harcourt, Brace & World, Inc.*

Many people have confused the scientific method with laboratories and test tubes. But a laboratory is only a room where there are special devices for turning up facts. Charles Darwin, who is regarded as one of the world's greatest scientists, did not use a laboratory. The world was his laboratory. He needed no special apparatus to uncover the facts that suggested and confirmed his theory of evolution. The scientific method is an attitude—a dogged insistence that no matter how right something sounds, we're going to check it out by observation. Sometimes this means tests, and sometimes it only means opening our eyes to observe facts that have been around us for years. So, if you want to boil down the method of science to one word, it's simply *testing*.

As man reaches out toward the twenty-first century, he will learn to be suspicious of all ideas that are not formulated so that they can be tested by observation. He will realize that the history of human thought shows that the ideas of which we are surest are the ones we most need to test. He will realize that his common sense only mirrors his training and experience. What seems natural and right to him is usually a reflection of the conditions under which he spent his first decade of life.

New generations, who will live and breathe the scientific spirit, will supplant us. Prejudice, grasshopper-like guessing, and emotional thinking will be rare. People of the future will, as suggested by John Dewey, achieve, ". . . the habit of suspended judgment, of skepticism, of desire for evidence, of appeal to observation rather than sentiment, discussion rather than bias, inquiry rather than conventional idealizations." They will know when further logical manipulation is fruitless. They will know when to stop discussing and check the facts. They will be like the proverbial "Man from Missouri"—show me. If they want to know whether a pudding's good, they won't just read the recipe;

they know that "the proof of the pudding is in the eating." They will feel, as Karl Pearson does, that, "There is no short cut to truth, no way to gain a knowledge of the universe except through the gateway of scientific method."

Attitudes that Help Us Develop Reliable Knowledge

In his book How to Develop Your Thinking Ability one of the authors has described in detail how the scientific method of thought can be used in meeting everyday life problems.*

* Kenneth S. Keyes, Jr., How to Develop Your Thinking Ability (New York: McGraw Hill Publishing Co., 1950).

Men, women, and children in the twenty-first-century world will probably use these or similar techniques to make the scientific spirit a way of life, not just something they use every now and then. They will hold ideas tentatively, not as bastions to be defended, but as tools to be improved. They will keep their eyes and minds open to find facts that do not support their points of view, for contrary facts may lead them to formulations that have greater predictability.

Because of limitations in our sensory and cortical equipment, no one can know all about anything in this world. All of our senses are limited in range. Dogs can hear higher-pitched sounds than we can. Our eyes can see only a small part of the electromagnetic spectrum. Edison said, "We don't know one-millionth of one per cent about anything." Since we can't know all there is to know about anything we must always keep an open mind for important factors that have been left out of our thinking. Wendell Johnson pointed out, "An attitude of this kind—'You can't tell me anything about that'—has an effect quite similar to that of a pus sac in the brain."

The world in which we live is constantly changing. No object in this world is without change. "The world rolls," said Ralph Waldo Emerson, "the circumstances vary every hour." On the atomic level all we have is motion and dynamism—perpetual split-second change. If we would act in ways that are effective and bring us the most happiness, we must train our senses to scan the world around us constantly to detect things that may have changed in a significant way. Alfred North Whitehead said, "Knowledge keeps no better than fish."

Another thing that helps us achieve the scientific spirit is to remember that no two things in this world are absolutely identical. Two things may be similar for our purpose, but the closer we look, the more differences we find. We become prejudiced when we lump a group of people under a single label and then respond to individuals as though they have the same characteristics as the label. Only open eyes and open minds are prepared to cope with a world in which no two things are alike. The words we use imply similarity. We must use our eyes and ears to remind us of differences that are important for our purpose.

Men, women, and children in the twenty-first century will learn to think in terms of degrees. The language we use often implies polar opposites—good or bad, true or false, beautiful or ugly, fast or slow, black or white. But the world in which we live usually shows a large number of degrees between extremes.

If we are to be as relaxed and happy as possible, our thoughts must adequately reflect the reality around us. And we can't do that by making black and white statements if the area to which we are referring contains shades of grey.

Individuals in the twenty-first century will learn to think in terms of probability. They will realize that man must regard all his knowledge as more or less probable. "Absolute certainty," said C. J. Keyser, "is a privilege of uneducated minds—and fanatics. It is, for scientific folks, an unattainable ideal." The people of the future will think of their ideas in terms of an ascending scale of probability, ranging from, "This seems most unlikely," through, "This may be or may not be confirmed by further observation," to, "This has a very high degree of probability."

When people adapt their thinking to the degree nature of our world, they will be more relaxed. They will be

more effective at locating and adopting, but always tentatively, the points-of-view that best represent the world about them. "A truly scientific attitude," said Dr. Roger Williams, "is one of humility. . .

A know-it-all attitude is incompatible with the scientific method." individuals in the twenty-first century will be acutely aware of the way their own nervous system influences their observations and reactions. We see life through the filter of our own individual personality and mode of thinking. Even the language structure that we absorb plays an important part in how we think and the way we observe things. Our ego-needs play a big part in selecting what we notice, fail to notice, remember, or forget. "We see things not' as they are," said the wise man, "but as we are."

Individuals in the twenty-first-century world will have a pro found feeling of the way all people and all things interact with their environment. People or things are not cut-and-dried entities. The way they act varies depending on the time and place. We must notice differences. Wendell Johnson said, "To a mouse, cheese is cheese. That is why mouse traps are effective."

Ever Lovelier Worlds

The success of the method of science in solving almost every problem put to it will give individuals in the twenty-first century a deep confidence in its effectiveness. They will not be afraid to experiment with new ways of feeling, thinking, and acting, for they will have observed the self-corrective aspect of science. Science gives us the latest word, not the last word. They will know that if they try something new in personal or social life, the happiness it yields can be determined after sufficient experience has accumulated. They will adapt to changes in a relaxed way as they zigzag toward the achievement of their values. They will know that there are better ways of doing things than have been used in the past, and they will be determined to experiment until they have found them. They will know that most of the unhappiness of human beings in the mid-twentieth century was not due to the lack of shiny new gadgets; it was due, in part, to not using the scientific method to check out new political and social structures that could have yielded greater happiness for them.

About a century ago Abraham Lincoln brilliantly expressed the attitudes that will most effectively help us work toward a happier future: "The dogmas of the quiet past are inadequate to the stormy present. The occasion is piled high with difficulty, and we must rise with the occasion. As our case is new, so we must think anew and act anew."

Future generations of mankind will realize that it is only through the scientific method of thinking that their value systems can be fully realized. They will welcome experimentation of all kinds in all phases of life. They will have a habitual open-mindedness coupled with a rigid insistence that all problems be formulated in a way that permits factual checking. They will have the attitude described by Wendell Johnson, "To a scientist a theory is something to be tested. He seeks not to defend his beliefs, but to improve them. He is, above everything else, an expert at 'changing his mind.' "*

* Wendell Johnson, People in Quandaries (New York: Harper & Brother 1946), p. 39. '

The paramount role that the method of science will play in helping us achieve "Life, Liberty, and the Pursuit of Happiness" was eloquently expressed by Herman J. Müller:

> *Above all, the spirit of science is the spirit of progress. . . .*
> *It can afford men ever newer horizons and higher peaks to climb, materially, mentally, and spiritually. It can afford ever greater and more inspiring opportunities for cooperative as well as individual achievement. Its pathway leads not only outwards into space and to other worlds than ours, but also inwards into the recesses of life, of the mind, and of the heart. By its means we will ourselves assume the role of creators of ever lovelier worlds and more sublime beings.***
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> ** Herman J. Müller, ". . . Therefore Choose Life" (Santa Barbara, California: Center for the Study of Democratic Institutions, 1965), p. 37.

6.Cybernated Technology

How would you like to have a guaranteed life income of $100,000 per year—with no taxes? And how would you like to earn this income by working a three-hour day, one day per week, for a five-year period of your life, providing you have a six-months vacation each year? Sound fantastic? Not at all with modern technology. This is not twenty-first-century pie-in-the-sky. It could probably be achieved in ten years in the United States if we applied everything we now know about automation and computers to produce a cybernated society. It probably won't be done this rapidly, for it would take some modern thinking applied in an intelligent crash program. Such a crash program was launched to develop the atomic bomb in a little over four years.

It might otherwise have taken thirty years. We get moving if we're threatened, but we chew the fat a long time when it comes to constructive improvements in human affairs.

You will recall that there are three factors that will play a vital role in the evolution of our civilization. One is our value system. Another factor is our method of thinking. The third is the state of our technology—the methods and machines for producing goods and services. The technological wave of the future will involve automated machines guided by computers.

When Queen Julianna of the Netherlands saw a demonstration of an electronic computer at an Amsterdam exhibition, she said, "I can't understand it. I can't even understand the people who can understand it." But the situation isn't really as bad as the queen implies. It isn't necessary to understand how a computer works any more than it is necessary to understand how the internal combustion engine in your car works for you to enjoy the benefits of the automotive age. The important thing is that we understand the effects of automation and computers. And that's what this chapter is about.

"The electronic computer," says Dr. Louis T. Rader, a vice-president of General Electric, "may have a

more beneficial potential for the human race than any other invention in history, Sir Leon Bagrit, who is the head of Britain's Elliot-Automation, has said that the computer and automation will bring, "the great est change in the whole history of mankind." Let's find out what they're talking about.

Automation simply means replacing human hands and feet by machines that do the same job—only better. Computers today replace human brains with electronic equipment that manipulates figures, makes programmed decisions, and gives instructions far more efficiently than any human. Cybernation means the control of the entire factory by a computer that acts in place of the boss.

Development of Automation

The purpose of machines is to lighten the burden of work. Suppose there were no machines. In such a society a person might have to work from 100 to 200 days per year just to get enough food for himself. To produce even a shirt to cover his back could require over 100 hours of labor! Suppose you would have to produce a shirt under primitive conditions. How many hours would you spend preparing soil and growing the cotton? After the cotton bloomed you'd have to pick it, tease out the seed, and spin the fiber into thread. Once you have enough thread, you could weave it into cloth. Then you would need to cut the cloth into the form of a shirt and sew it together. Imagine the amount of time it would take if your only tools were a hoe, a knife, and a needle.

The time needed to produce a shirt took a nose dive back in the eighteenth and nineteenth centuries when much of the work became mechanized and concentrated in factories. The development of the cotton gin, which separated the seed from the white cotton fiber, the invention of machinery that made thread, and the design of looms that wove the thread into cloth enabled a better shirt to be manufactured with only several hours of human labor. Today a shirt may represent only a little over one hour of human time from seed to salesman!

In the twentieth century mass production with assembly lines and improved machinery greatly reduced the cost of producing goods. Automation is based on all of the principles of mechanization and mass production, but it goes one step further. Previously factories needed humans to operate each machine. Automated machines operate themselves. Through "feedback" mechanisms they observe what they are doing. They give themselves instructions, and they check on the quality of their output. They work faster and at speeds that would kill a human. They don't get tired and they don't forget. They never go out on strike or ask for pay increases. Factory design is simplified. Automatic machines don't need parking lots, air conditioning, bright lighting, wash rooms, lunchrooms, or coffee breaks. They don't even need buildings for many types of work.

Automatic machines have been, or soon will be, designed to perform almost any conceivable task done by human beings. Since we have only two hands, it is possible to design automatic machines that will far exceed the manipulative ability of a human. In 1961 U.S. Industries announced that they had developed the first general-purpose automation machine at a price of around $2,500. It is called the TransfeRobot. Its swinging arm and hand is infinitely superior to any human arm or hand. It never gets tired, and the